

Entity-Relationship (ER) Diagram

The Ucab database relies on interconnected collections. The relationships define how users interact with drivers via rides.

- **User:** Stores passenger credentials, payment methods, and wallet balance.
- **Driver:** References a User document for authentication, but extends it with vehicle details, verification status, and geospatial location tracking.
- **Ride:** The core transactional entity linking a User and a Driver. It stores pickup/dropoff coordinates, fare details, and timestamps.
- **Payment:** Tracks the financial transaction related to a specific Ride.

ER Diagram

```
erDiagram
    USER ||--o{ RIDE : books
    USER ||--o| DRIVER : is_a
    DRIVER ||--o{ RIDE : accepts
    RIDE ||--|| PAYMENT : generates

    USER {
        ObjectId _id
        String name
        String email
        String role
    }

    DRIVER {
        ObjectId _id
        ObjectId user_id
        String vehicleType
        Point currentLocation
        Boolean isAvailable
    }

    RIDE {
        ObjectId _id
        ObjectId user_id
        ObjectId driver_id
        Point pickupLocation
        Point dropoffLocation
        String status
        Object fare
    }

    PAYMENT {
        ObjectId _id
        ObjectId ride_id
        Number amount
    }
```

String status

}